Lab Assignment 6

Title: Implement a hash table of size 10 and use the division method as a hash

function. In case of

a collision, use chaining.

Program :

```
#include <iostream>
#include <list>
using namespace std;
const int SIZE = 10; list<int>
table[SIZE];

int hashFunction(int key) {
    return ((key % SIZE) + SIZE) % SIZE;
}

void insertKey(int key) {
    int index = hashFunction(key);
    table[index].push_back(key);
    cout << "Key inserted successfully!";
}

void searchKey(int key) {
    int index = hashFunction(key);

    for (int value : table[index]) { if
        (value == key) {
            cout << "Key found at index " << index; return;
        }
    }
```

```
    cout << "Key not found!";
}

void displayTable() {
    for (int i = 0; i < SIZE; i++) { cout
        << "\n" << i << " -> ";

        for (int key : table[i]) cout <<
            key << " -> ";

        cout << "NULL";
    }
}

int main() {
    int choice, key;

    do {
        cout << "\n\n1. Insert"; cout
        << "\n2. Search"; cout <<
        "\n3. Display"; cout << "\n4.
        Exit";
        cout << "\nEnter choice: "; cin
        >> choice;

        switch (choice) { case
            1:
                cout << "Enter key: ";
                cin >> key;
                insertKey(key); break;

            case 2:
                cout << "Enter key to search: "; cin >>
                key;
                searchKey(key); break;

            case 3:
                displayTable();
                break;

            case 4:
                break;
               default:
                cout << "Invalid choice!";
        }
    } while (choice != 4);
    return 0;

}
```

Output:

```
1. Insert
2. Search
3. Display
4. Exit
Enter choice: 1
Enter key: 1
Key inserted successfully!

1. Insert
2. Search
3. Display
4. Exit
Enter choice: 1
Enter key: 2
Key inserted successfully!

1. Insert
2. Search
3. Display
4. Exit
Enter choice: 1
Enter key: 3
Key inserted successfully!

1. Insert
2. Search
3. Display
4. Exit
Enter choice: 1
Enter key: 4
Key inserted successfully!

1. Insert
2. Search
3. Display
4. Exit
Enter choice: 2
Enter key to search: 4
Key found at index 4

1. Insert
2. Search
3. Display
4. Exit
Enter choice: 3

0 -> NULL
1 -> 1 -> NULL
2 -> 2 -> NULL
3 -> 3 -> NULL
4 -> 4 -> NULL
5 -> NULL
6 -> NULL
7 -> NULL
8 -> NULL
9 -> NULL

1. Insert
2. Search
3. Display
4. Exit
Enter choice: 4

Process returned 0 (0x0)   execution time : 37.011 s
Press any key to continue.
```